EXTRACELLULAR MATRIX

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LECTURE OF OBJECTIVES

- Describe extracellular matrix
- List the biomolecules of the ECM
- List the proteins of ECM
- Discuss the functions of ECM
- Justify that Structure and consistency of ECM changes with the location

- the **extracellular matrix** (ECM) is a collection of **extracellular** molecules secreted by cells that provides structural and biochemical support to the surrounding cells.

VARIATIONS IN ECM WITH THE SITE

- **Ground substance**
  - Between the cells
- **Basal Lamina**
  - Separating tissues (bowman capsule)
- **Tendon**
  - Joining tissues
- **Cartilage, bones**
  - Hard connective tissue

Composition of Extracellular Matrix (ECM)

- Cells (mesenchymal origin)
  - fibroblasts
  - smooth muscle cells
  - chondroblasts
  - osteoblasts and epitelial cells
- Organic fibrilar matrix
- Organic nonfibrilar matrix
- Water

FUNCTIONS OF ECM

- Ground substance
- Pathway for chemical modulators/ migratory cells (hel
  in cell differentiation and growth)
- Separates cells and tissues
- Mechanical connection between the cells (bones,
  cartilages)
- Filters in renal capsules

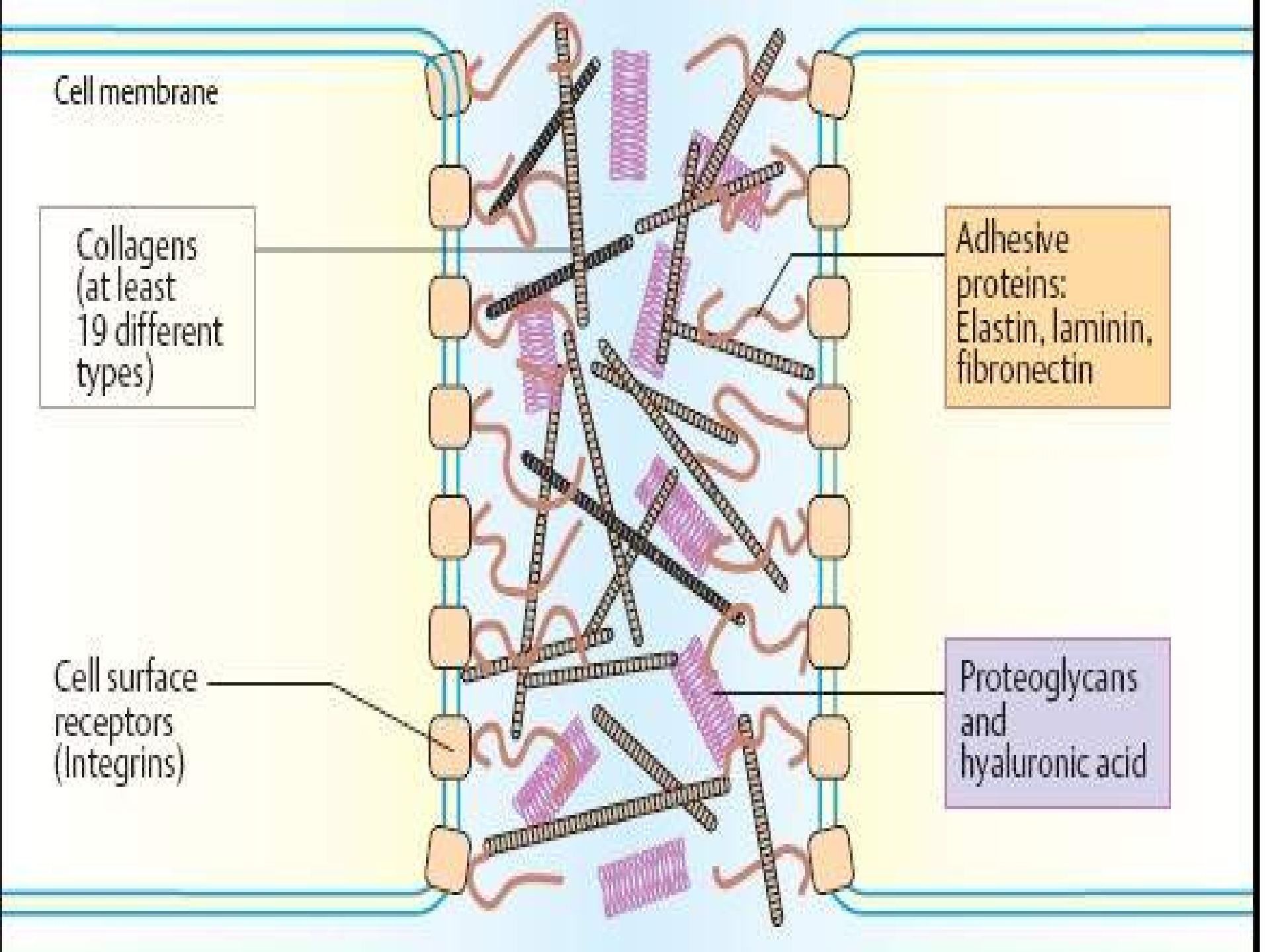
Structure of ECM

- **collagen**
  – the main ECM component, forms the main fibres

- **elastin**

- **proteoglycans**
  - heteropolysacharides

- **structural glycoproteins**
  - fibronectin, laminin


Cell membrane
Collagens (at least 19 different types)
Adhesive proteins: Elastin, laminin, fibronectin
Cell surface receptors (Integrins)
Proteoglycans and hyaluronic acid

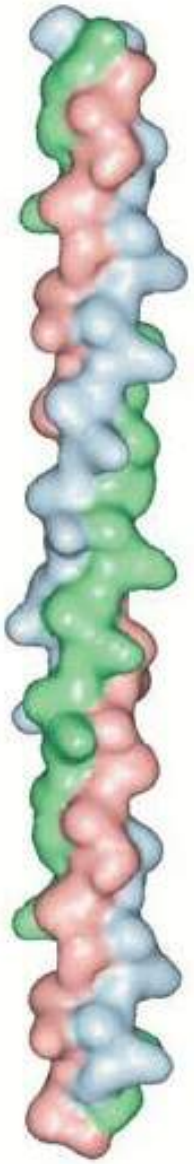
Collagen

- The most abundant protein in the body, making 25%-35% of all the whole-body proteins.
- Collagen contributes to the stability of tissues and organs.
- It maintains their structural integrity.
- It has great tensile strenght.
- The main component of fascia, cartilage, ligaments, tendons, bone and skin.
- Plays an important role in cell differentiation, polarity, movement

STRUCTURAL PROTEIN: COLLAGEN

- Triple helix
- 25 different types
- Varies with the site
- Repeated structure of three amino acids
  - $\{Glycine\text{-}X\text{-}Y\}_n$
- Fibres, fibrils, mesh works and ligaments
- Tensile strength and flexibility
- 30 genes encode the protein
- Ehler Danlos Syndrome
  - Hyperextensibility of skin
  - Abnormal tissue fragility
  - Increased joint mobility

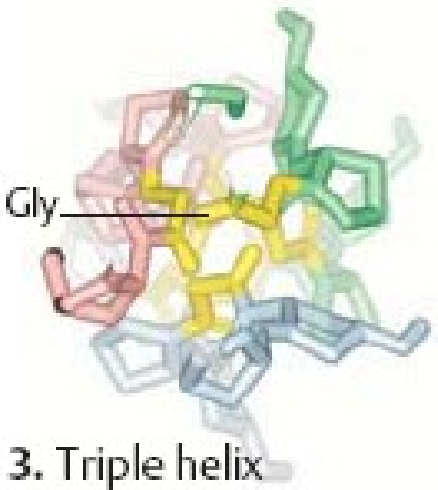
B. Collagen



1. Triple helix (section)

Gly	Arg	Hyp
Gly	Gln	Arg
Gly	Pro	Hyp
Gly	Pro	Gln
Gly	Ala	Arg
Gly	X	Y

2. Typical sequence



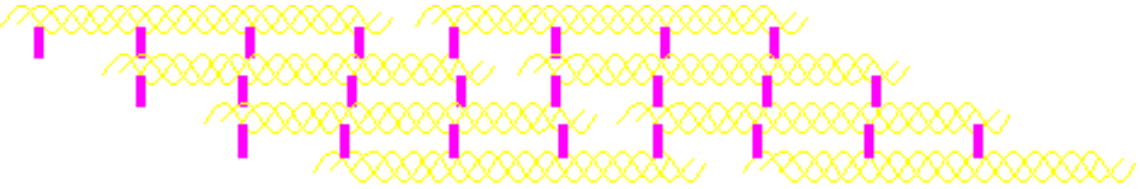
3. Triple helix
(view from above)

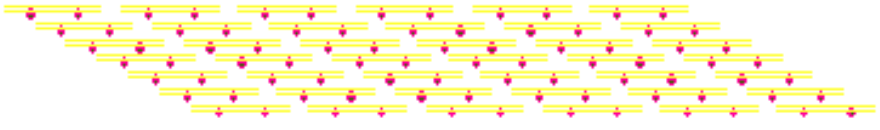
STRUCTURAL PROTEIN: COLLAGEN

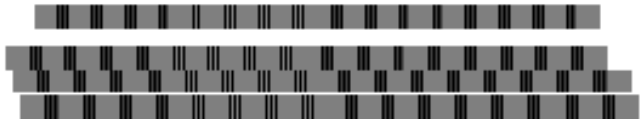
- 19 types identified (Harper's Biochemistry 29the ed Page Table 48-1)
- Considerable post-translational modifications
  - Preprocollagen – procollagen – collagen
  - Hydroxylation of prolyl and lysyl residues
  - Vitamin C coenzyme
- Inter and Intra chain crosslinks facilitate triple helix

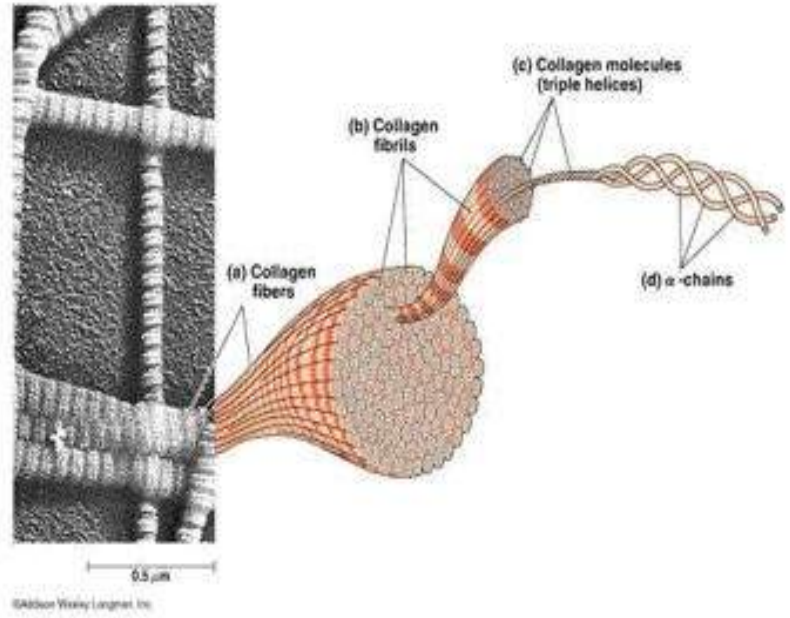
8. Self-assembly or polymerization of tropocollagen molecules form collagen fibrils.

Cross-linkage between adjacent tropocollagen molecules stabilizes the fibrils.

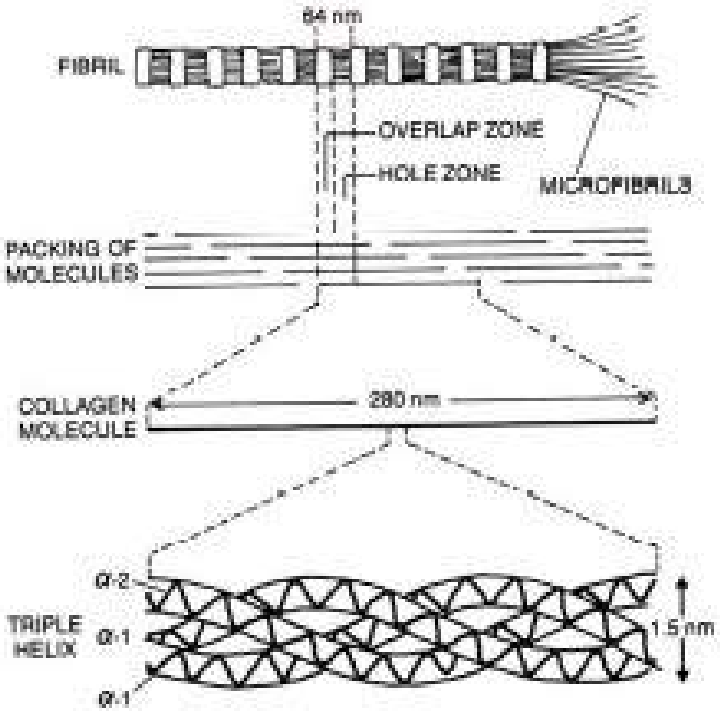




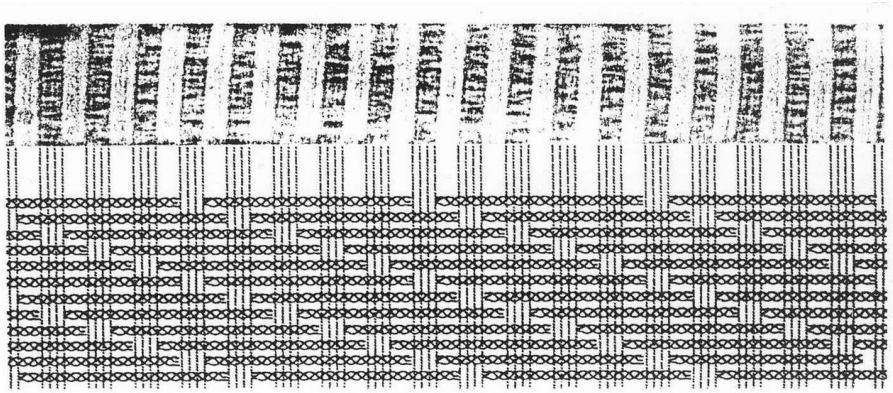


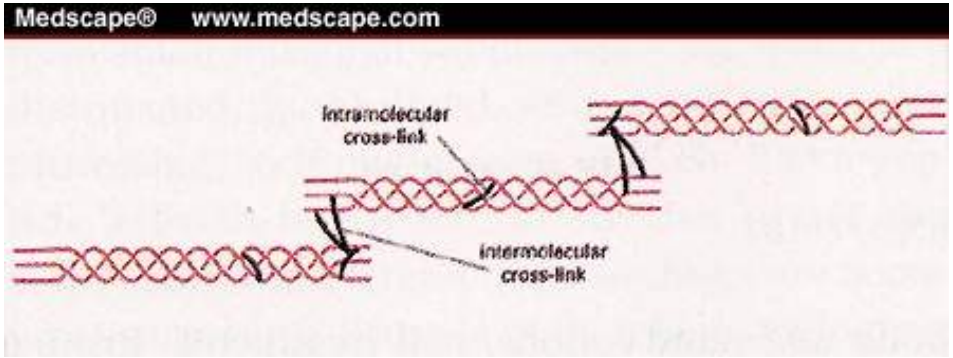


Collagen – fiber formation



Collagen types I, II, III, V, IX, X

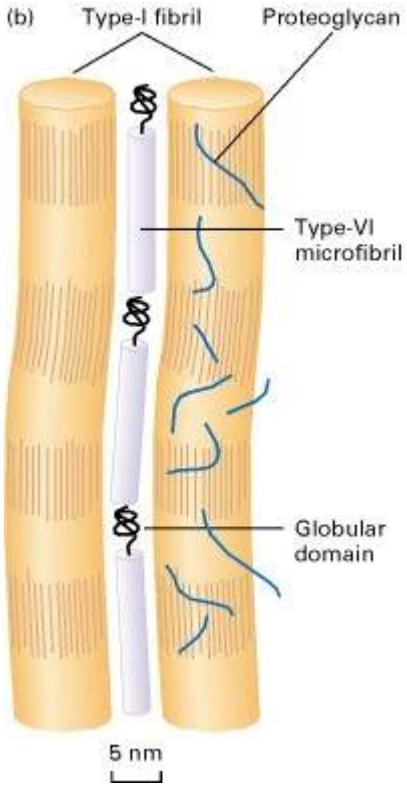




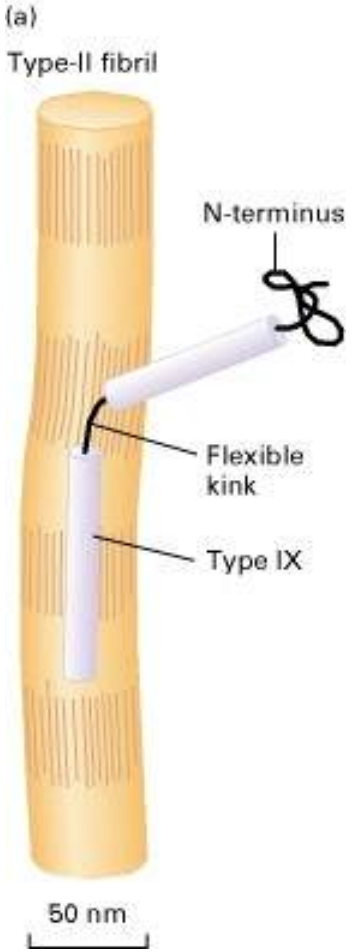
Cross striated structure in

Collagen Interactions

Tendon



Cartilagous matrix



Collagen Distribution

Type	Molecule composition	Tissue
Fibrilar Collagens		
I	$[\alpha1(I)_2\alpha2(I)]$	Skin, tendon,bone, ligaments, dentin
II	$[\alpha1(II)]_3$	Cartilage, vitreus humor
III	$[\alpha1(III)]_3$	Skin, muscle, blood vessels
V	$[\alpha1(V)]_3$	Similar to type I, fetal tissue
Sheet-forming collagen		
IV	$[\alpha1(IV)_2\alpha2(IV)]$	
	$[\alpha1(IV)\alpha2(IV)\alpha3(IV)]$	All basal laminaes

DISEASES OF COLLAGEN FIBRES

- **Ehlor Deanlos Syndrome**
  - Inherited
  - Defective hydroxylation reactions
  - Hyperextensibilty of skin
  - abnormal tissue fragility

DISEASES OF COLLAGEN FIBERS

- **Alport Syndrome**
  - Applied to inherited disorder involving defective Type IV collagen
  - Hematuria
  - End-stage renal disease
- **Epidermolysis Bullosa**
  - Type Vii collagen
  - Skin breaks and blisters
- **Scurvy**
  - Vitamin C deficiency
  - Defective Hydroxylation

STRUCTURAL PROTEIN: FIBRILLIN

- Glycoprotein
- Secreted form fibroblast
- Incorporated in microfibrils
- Provide scaffold for deposition of elastin

MARFAN SYNDROME

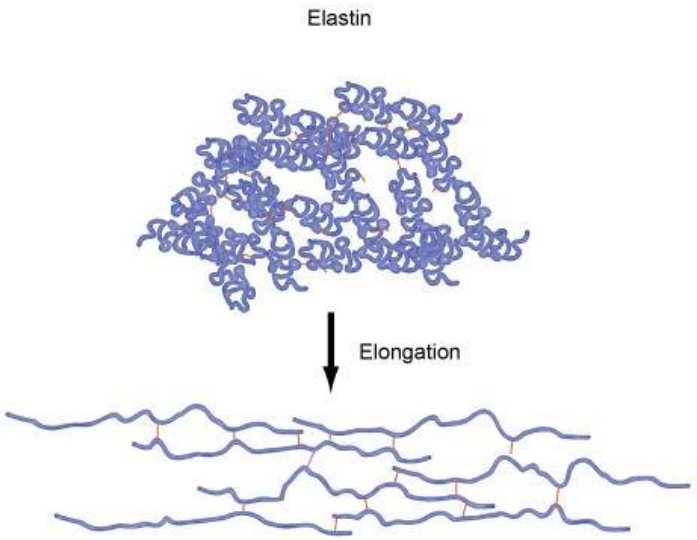
- Mutations in fibrillin genes
- Autosomal dominant
- **Eyes**: ectopia lentis
- **Skeletal system**:
  - Tall with arachnodactyly and hyperextensibilty
- **Cardiovascular system**: weakness of aortic media leading to dilation of aorta

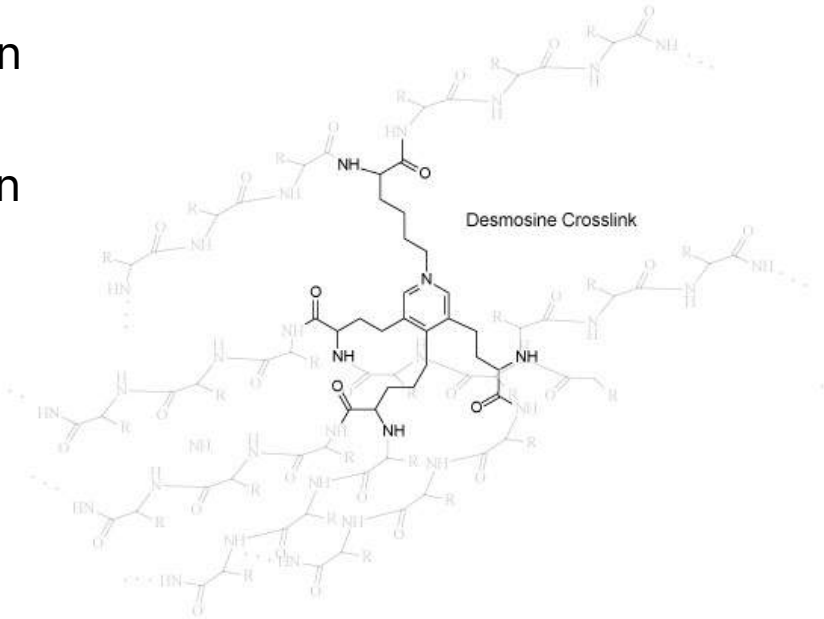
STRUCTURAL PROTEIN: ELASTIN

- Extensibility
- Elastic recoil
- Random coil confirmation permitting stretching
- Lungs, large arteries, elastic ligaments skin and ear cartilage
- **Elastin gene depletion** → Williams syndrome
  - Supravalvular aortic stenosis
- **Scleroderma** → accumulation of elastin

Elastin

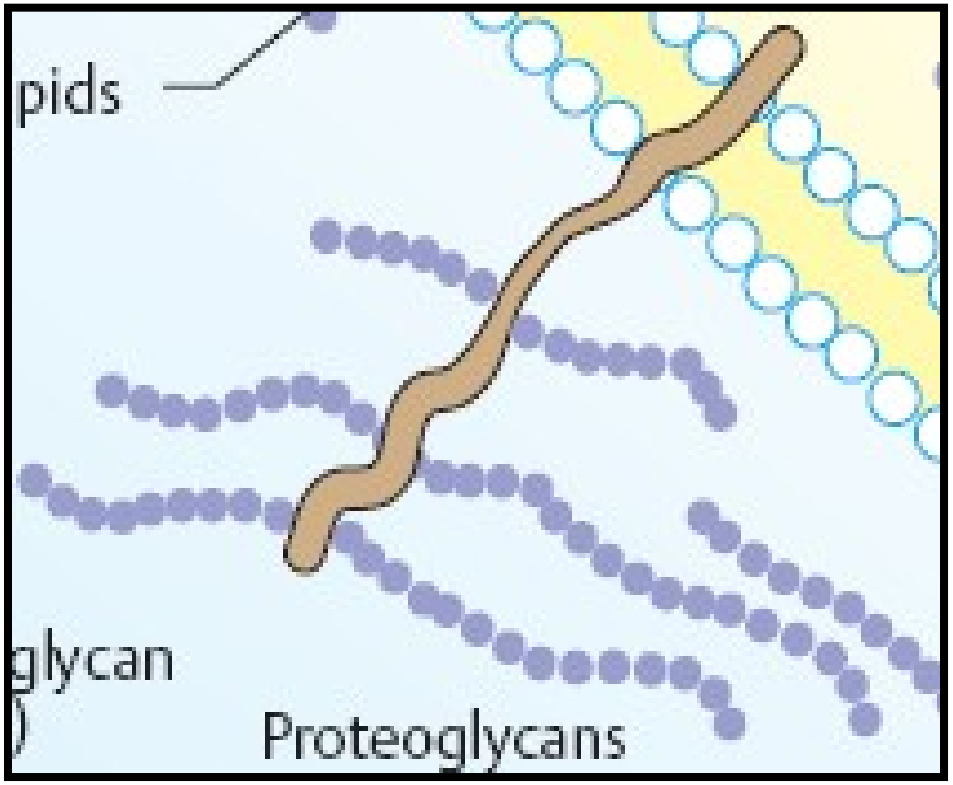
- Elastin is a major protein component of tissues that require elasticity such as arteries, lungs, bladder, skin and elastic ligaments and cartilage.
- It is composed of soluble tropoelastin protein containing primarily, glycine and valine and modified alanine and proline residues.
- Tropoelastin is a ~65kDa protein that is highly cross-linked to form an insoluble complex.
- The most common interchain cross-link in elastins is the result of the conversion of the amine groups of lysine to reactive aldehydes by lysyl oxidase. This results in the spontaneous formation of desmosine cross-links.





GLYCOSAMINOGLYCANS

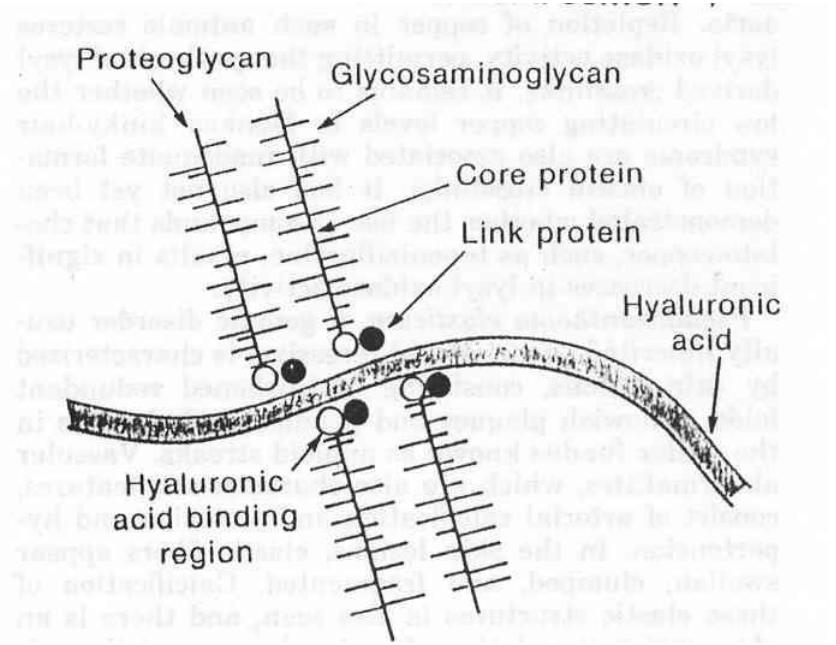
- Present in ECF
- Waterbinding
- Anticoagulant ICF
- List
  - Hyaluronic acid
  - Heparan sulfate
  - Heparin
  - Keratan sulfate
  - Chondroitin Sulfate



Proteoglycans

Proteoglycans represent a special class of glycoproteins that are heavily glycosylated (95%).





They consisit of core protein with one or more attached glycosamino glycan chain(s).

Glycosaminoglycans (GAG)

Glycosaminoglycan (GAG) chains are long, linear carbohydrate polymers
under physiological conditions they are negatively charged (due to the occurrence of sulfate and uronic acid groups).

Disaccharide subunits are:

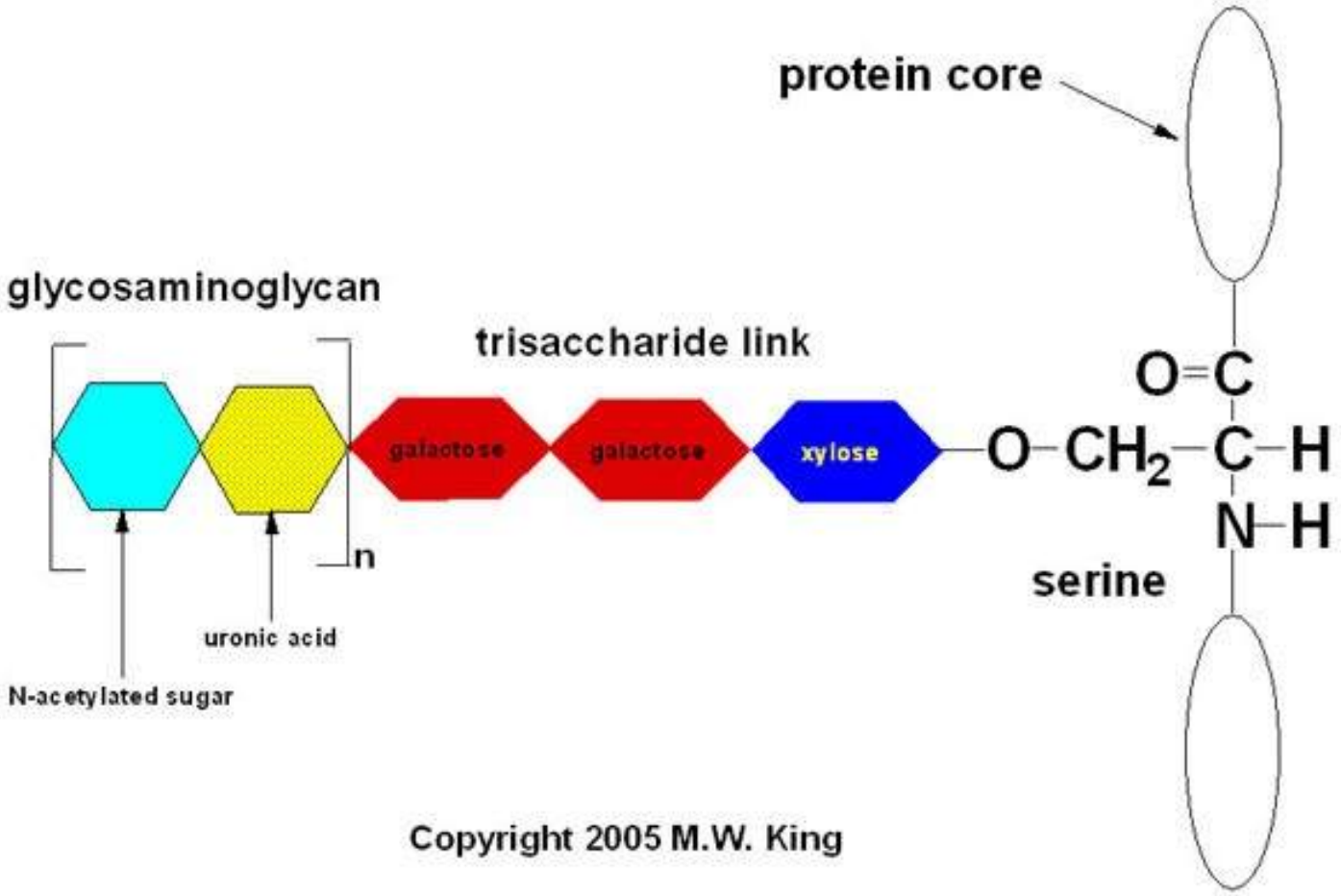
1. uronic acid

D-glucuronic acid or L-iduronic acid

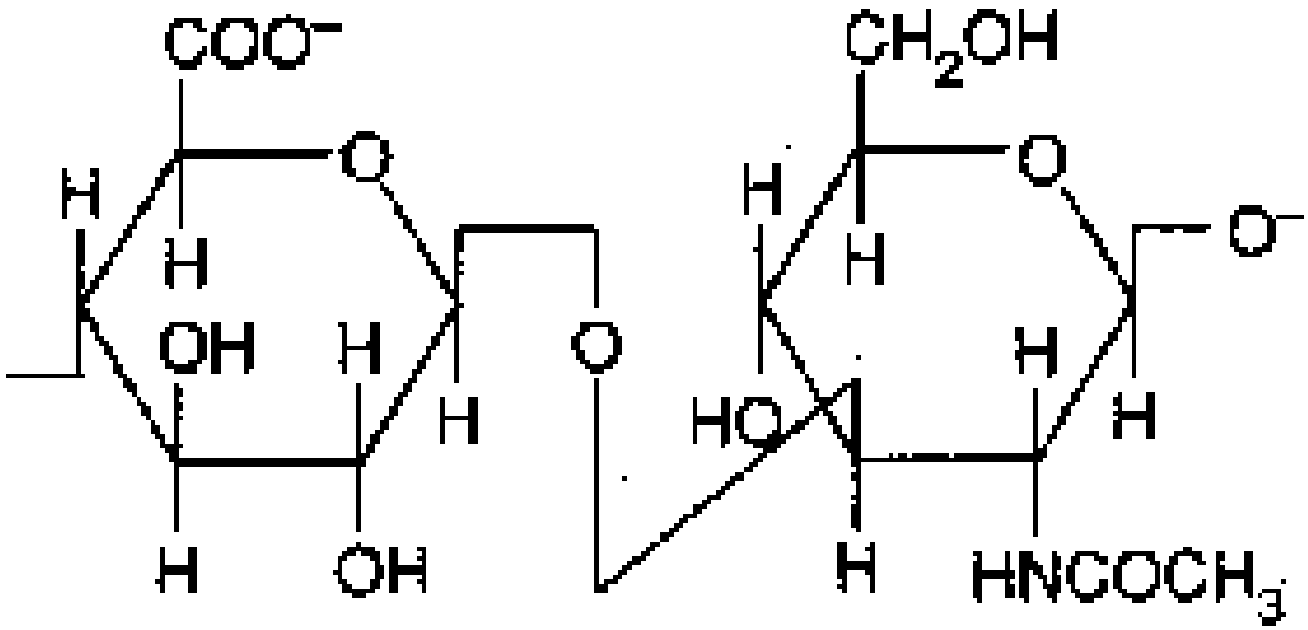
2. aminosugar

N-acetyl glucosamin (GlcNAc) or
N-acetyl galactosamin (GalNAc)

Linkage of GAGs to protein core by specific trisaccharide linker



Hyaluronic acid



D-glukuronic acid + GlcNAc

Synthesis

- The protein component is synthesized by ribosomes and transocated into the lumen of the RER.

- Glycosylation of the proteoglycan occurs in the Golgi apparatus i **multiple enzymatic steps**.

- First a special link tetrasaccharide is attached to a **serine side chain** on the core protein to serve as a primer for polysaccharide growth.

- Then sugars are added by *glycosyl transferase*. The completed proteoglycan is then exported in **secretory vesicles** to th extracellular matrix of the cell.

Glycosaminoglycan Classification

Proteoglycans can be categorised depending upon the nature of their glycosaminoglycan chains.

- **Hyaluronic acid** (does not contain any sulfate)
  - non-covalent link complex with proteoglycans
- **Chondroitin sulfate** cartilage, bone
- **Dermatan sulfate** skin, blood vessels
- **Heparan sulfate** basement membrane, component of cells surface
- **Keratan sulfate** cornea, bone, cartilage often aggregated with chondroitin sulfate

Function of Proteoglycans

- organize water molecules

occupy space between cells and collagen

- high viscosity
  - lubricating fluid in the joints
- specific binding to other macromolecules
- link to collagen fibers

cell migration and adhesion
  - passageways between cells
- anchoring cells to matrix fibers

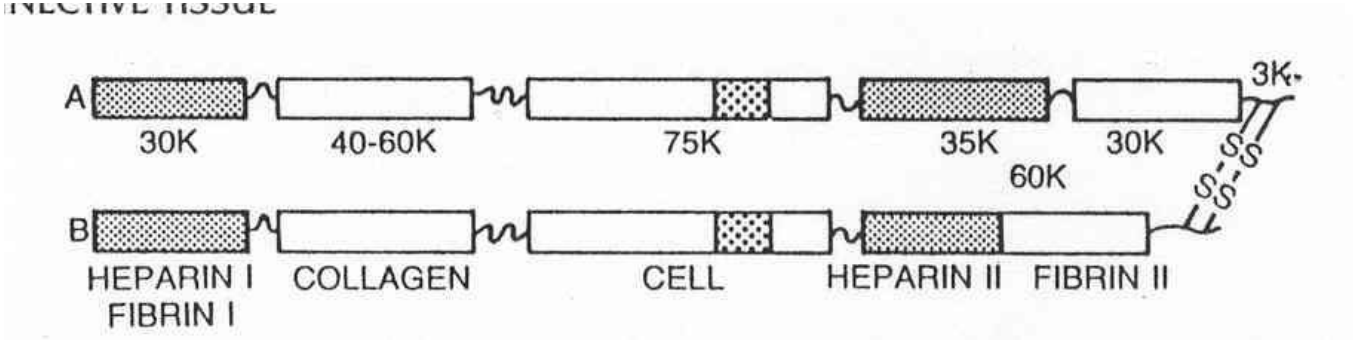
Structural Glycoproteins

- Direct linkage to collagen or proteoglycans
  - anchoring collagen fibers to cell membrane
  - covalent attachment to membrane lipid

- Major adhesive structural glycoproteins
  - **fibronectin**
  - **laminin**

Fibronectin Structure

- Dimer connected at C-terminal by S-S linkage
- Rigid and flexible domains
- Cell binding domain RGDS
  (arg, gly, asp, ser)
  - binding receptor in cell membranes
- Domain is binding to
  - collagen type I, II and III
  - heparin sulfate
  - hyaluronic acid
  - fibrin



Fibronectin Function

- cell adhesion
- cell differentiation
- anchoring basal laminae to other ECM
- blood clothing
  - clothing process, link to fibrin

Laminin Structure and Function

- cross-shaped glycoprotein
- 3 polypeptide chains
- domain bind
  - collagen type IV
  - heparin
  - heparin sulfate
- cell surface receptor
- cell adhesion
- cell differentiation
- anchoring the glycoprotein to basal laminae

